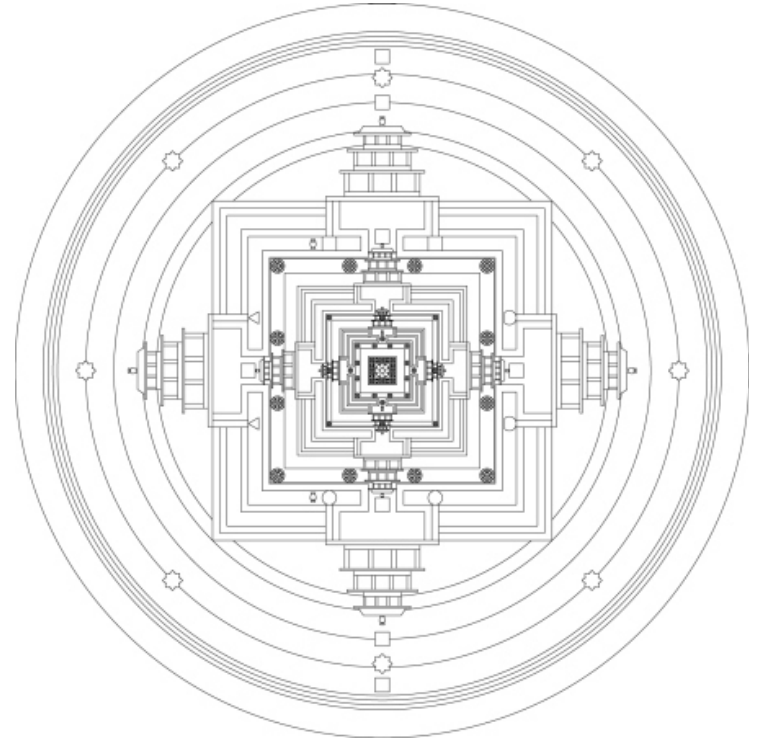


The Kālacakra maṇḍala according to the Vajrāvalī of Abhayākaragupta

1. Drawing the maṇḍala



This article is in two parts. This present page covers the drawing of the maṇḍala, while the [second page](#) describes the emblems of the deities. A PDF file of this article is available [here](#). An [associated page](#) describes the perimeter beings of the maṇḍala as used in two of the Tibetan traditions. (Such detailed lists are not given in the original Vajrāvalī.)

Perhaps the most important of the original Indian sources on the drawing of the Kālacakra maṇḍala, and one that is important to all the Tibetan traditions, is the Vajrāvalī-nāma-maṇḍalopāyikā (*dkyil 'khor gyi cho ga rdo rje phreng ba*) of Abhayākaragupta, an abbot of Vikramaśīla monastery, probably in the late 11th or very early 12th century CE. This text gives the details of how to prepare for the purpose of giving empowerment, 26 maṇḍalas.

Many Tibetans have elaborated on the methods given in the Vajrāvalī, and the tradition is very much still alive, particularly the transmission of the Kālacakra, although some frown on the Vajrāvalī Kālacakra because it allows the use of a painted rather than powder maṇḍala. This is considered by critics to be improper.

This article will describe the method of drawing the maṇḍala according to the Vajrāvalī and is based on three main sources: the most important of these is the Vajrāvalī itself, but I shall also refer to two Tibetan descriptions of the Vajrāvalī method. One of these is from the Jonang tradition, and is given by Banda Gelek in his text "The Illuminating Sun rays" (*rje sgrol ba'i mgon pos mdzad pa'i dkyil chog rgya mtsho'i thig tshon gsal byed nyi ma'i 'od zer*). The other is in the Gelug tradition, by Akhuching Sherab Gyatso: *"bla ma rdo rje 'chang chen pos dkyil chog rdo rje phreng ba sogs kyi mtshon pa'i nyer mkho'i thig dang bshad pa phyag len du ma'i skor la brjed byang mdzad pa"*. Any particular comments made by Sherab Gyatso, and there are several, are indicated by "SG:"

Units and dimensions.

The description starts by defining the overall dimensions of the maṇḍala and the units to be used in its construction. The first thing that needs to be done is to draw the central (*tshang* *thig*, brahmasūtra) and diagonal lines (*zur thig*, koṇasūtra). In Abhaya's description, these lines only extend to the inner bounding circle of the perimeter of earth. There are six such perimeters surrounding the maṇḍala, that of earth being the innermost. In practice, these main lines need to be drawn longer than this, but taking Abhaya's definition, each of these lines, measured from the centre to the inner circle of the earth perimeter, is eight cubits. The size of the maṇḍala is stated to be four cubits – this is the distance between the inside of the walls of the body palace.

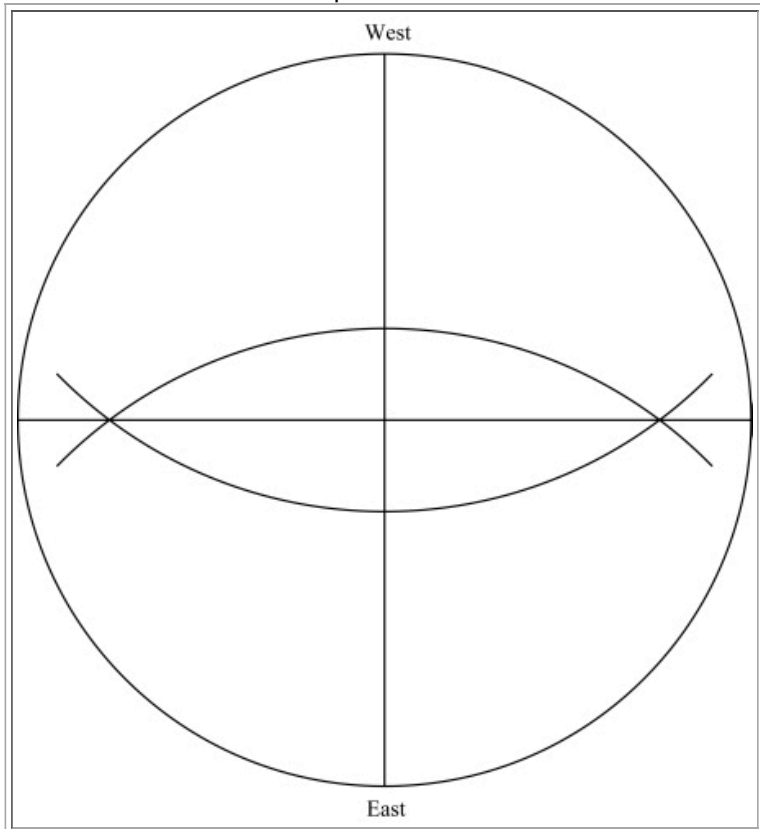
The Vajrāvalī has been analysed by Masahide Mori in his 1997 SOAS thesis, "The Vajrāvalī of Abhayākaragupta". This text is available to download free from the British Library's Electronic Theses Online Service, [EThOS](#). Registration (at no cost) is required.

Perhaps of more interest to those familiar with Tibetan and Sanskrit is the two-volume "Vajrāvalī of Abhayākaragupta : edition of Sanskrit and Tibetan versions", by Masahide Mori, published by The Institute of Buddhist Studies, Tring, UK, 2009.

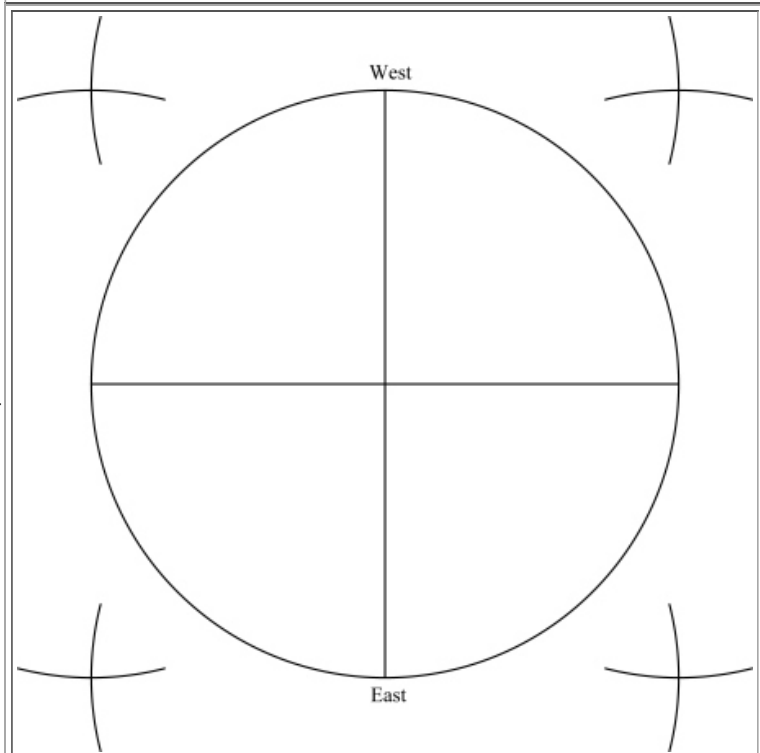
In a ritual context, the width of the presiding teacher's thumb is determined and 24 such finger-widths is one cubit. Half of one of these finger-widths is called a minor unit (*cha chung*); these are the main units used in describing the maṇḍala. Other units that are sometimes used are door units (*sgo tshad*, DU). The maṇḍala consists of three palaces, and the door unit for any palace is the width of the doorway of that particular palace. The mind palace is the smallest of the three, and its door unit (mind-DU) is equal to six minor units (mu). The speech palace is twice the size, and so one speech-DU is equal to 12 mu. The body palace is again twice the size, and so one body-DU is equal to 24 mu. The inner widths of the three palaces are, respectively: 48, 96 and 192 mu. Incidentally, in most other maṇḍalas, such as all the other maṇḍalas described in the Vajrāvalī, a minor unit is defined as one quarter of a door unit.

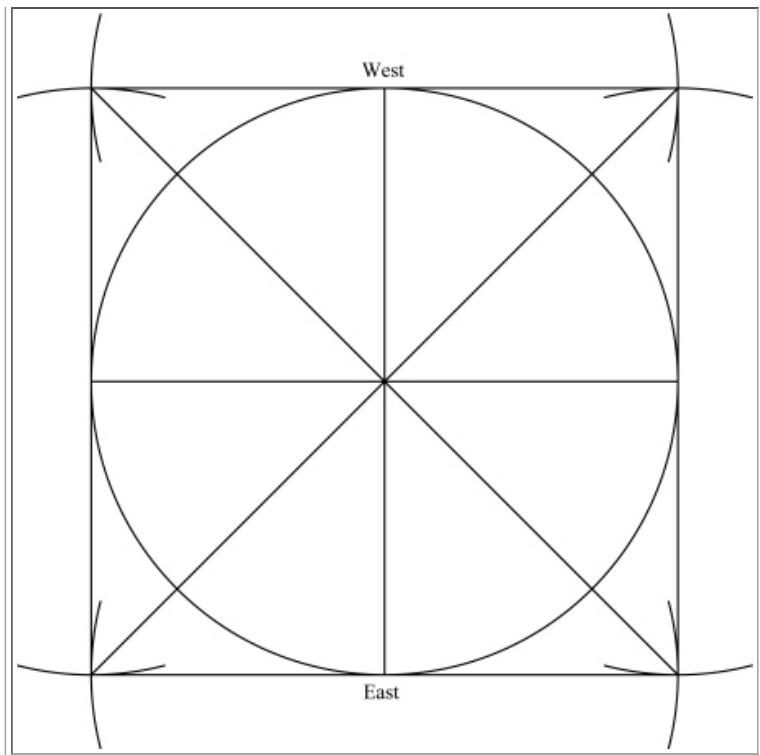
The first part of the construction is to draw the main lines, and then start measuring out the position of the walls, and so forth. For this initial description I shall follow Banda Gelek.

Having prepared the surface on which the maṇḍala is to be drawn, first, draw the east-west central line, and having found its centre, describe from that centre a circle of radius 8 body-DU. This is 192 mu, and forms the innermost boundary of the earth perimeter. From where that circle meets the central line, draw intersecting arcs of larger radius. Where these arcs cross draw the second central line, running north-south. These steps are illustrated in the diagram on the right.

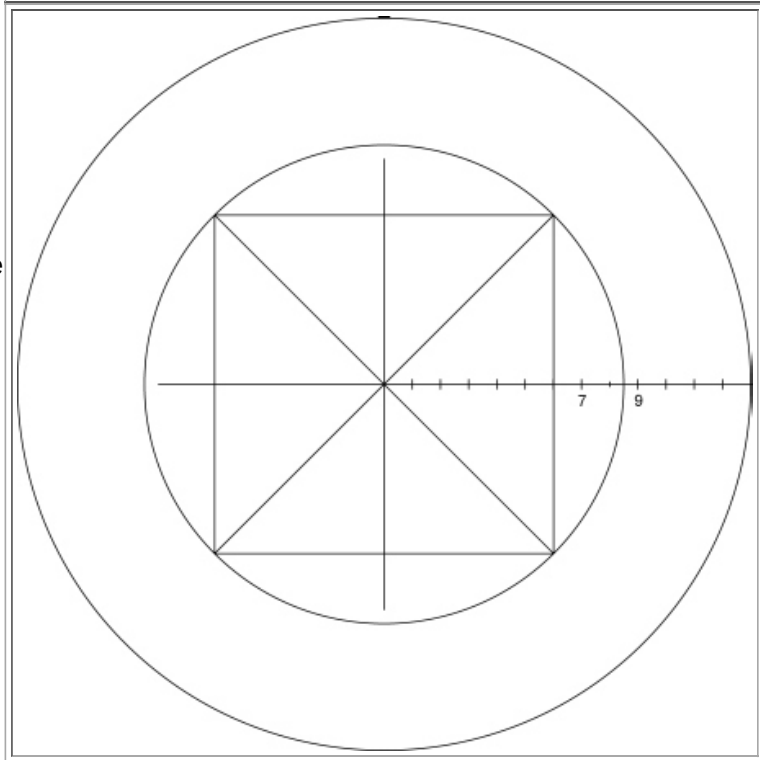


Then, where the two central lines meet the original circle, draw further intersecting arcs using the previous radius of 8 body-DU. Connect these together in order to form the two diagonal lines. In practice, one would first need to measure out the total area that is to be used for the maṇḍala. If the diameter of the inner bounding circle of the earth perimeter is 384 mu (2×192), that of the outer bounding circle of the outermost perimeter is 624 mu.



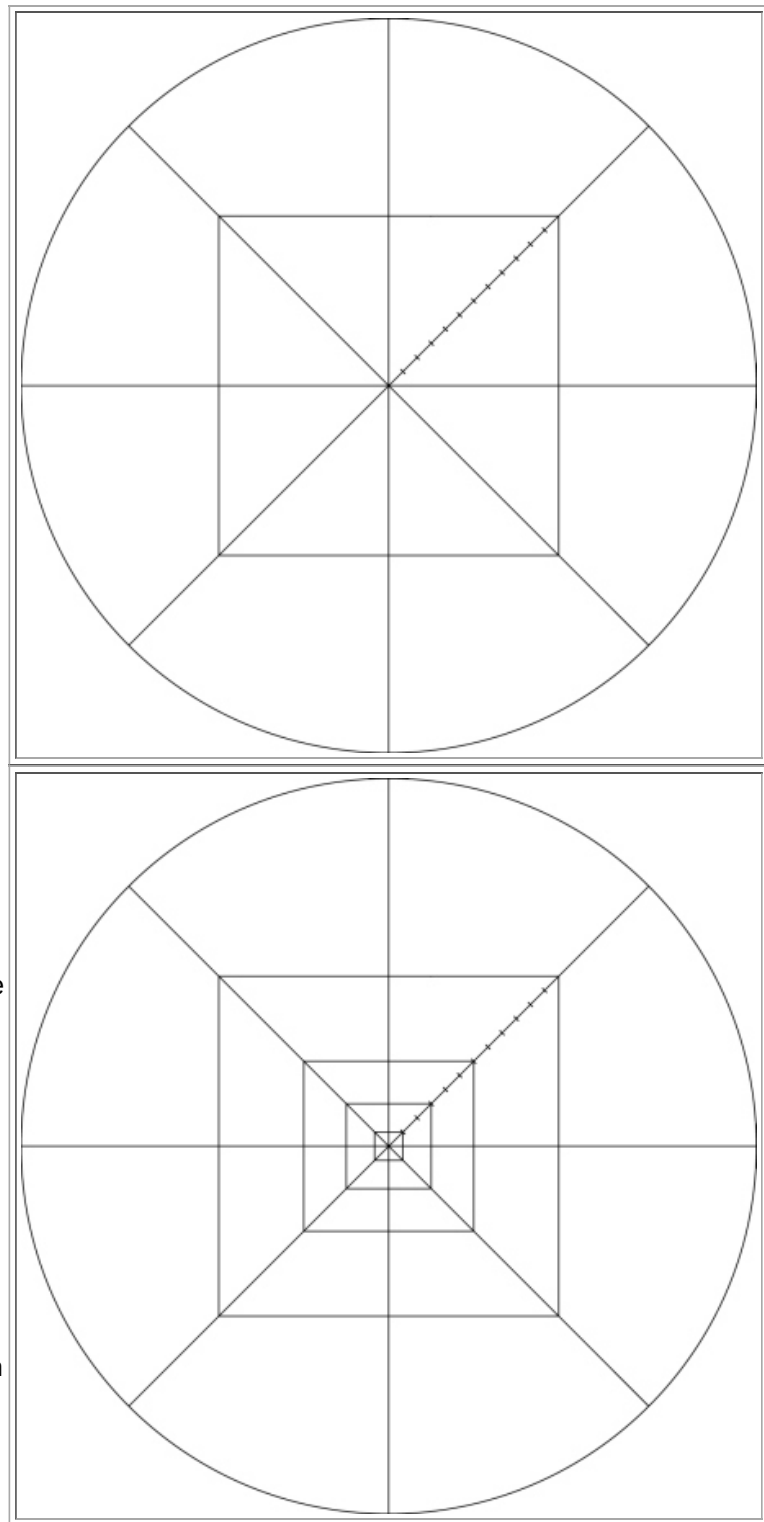


Banda Gelek's description takes the distance from the centre to the extreme edge, 312 mu, and divides this into 13 equal parts. Each of these is 24 mu, or 1 body-DU. Next, draw a circle with a radius of $8\frac{1}{2}$ body-DU (204 mu) and connect the points where this circle intersects the diagonal line to form a square. Clearly, the diagonal lines and at least one central line need to be longer than the standard 8 body-DU. The lines forming this square are the parapet lines (*mda' thig*, *kramaśīrṣasūtra*) for the body maṇḍala, and should each be 288 mu. There is actually a very small error in creating the square this way, but that error is small enough to ignore; if accurately drawn, the sides of the square will be 288.5 mu. (By Pythagoras' theorem: $204^2 \div 2 = 144.25^2$.) However, this method helps ensure a properly square shape.



The next step is to divide the diagonal lines from the corners of this square to the centre into 12 equal parts.

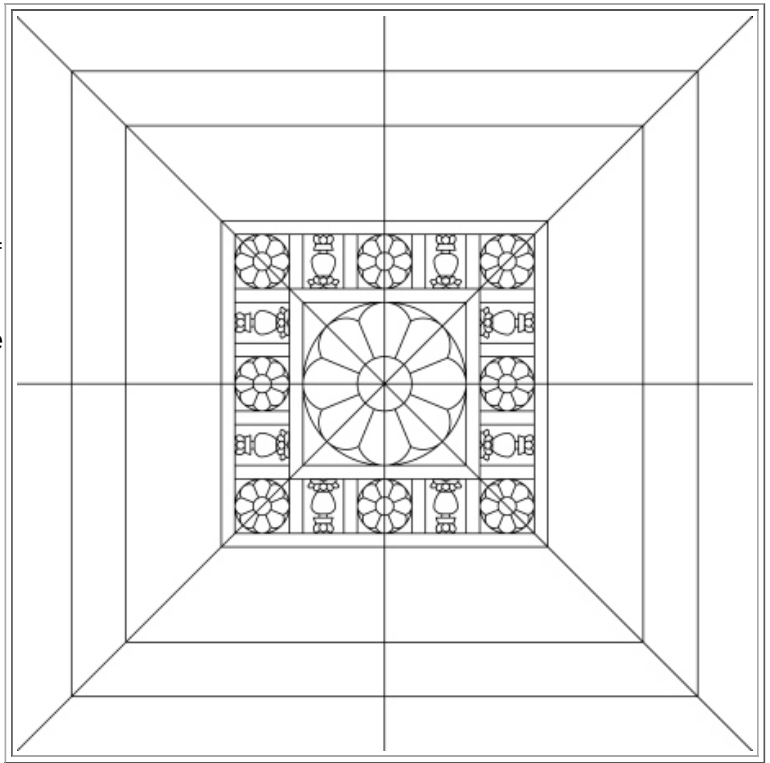
Draw further squares by connecting the first, third and sixth positions from the centre. From the centre, we now have four squares forming: the outer line of the outer beams of the circle of great bliss, and the parapet lines of the mind, speech and body palaces. All these squares are final lines and not construction lines, and so they will not need later to be erased. According to Banda Gelek's interpretation, the method of the Vajrāvalī is intended to do away with construction lines; the descriptions in the Tantra and Vimalaprabhā require many to be drawn, if taken literally. However, it is impossible to do away with construction lines completely. These steps are illustrated in the diagrams on the left



Measuring out from the centre, first a circle is drawn with radius 2 mu; this is for the receptacle of the central lotus. Then, measure out along the central lines, 4, 1, 4 and 1 mu, drawing squares, again with their corners on the diagonal lines. The 1 mu spaces between the first two of these squares and the outer pair form the inner and outer beams of the circle of great bliss. Abhaya refers to these as vajra-garlands, after the design that will later be drawn on them. Another circle also needs to be drawn inside the inner beams, of radius 6 mu. This forms the space for the eight petals of the main lotus.

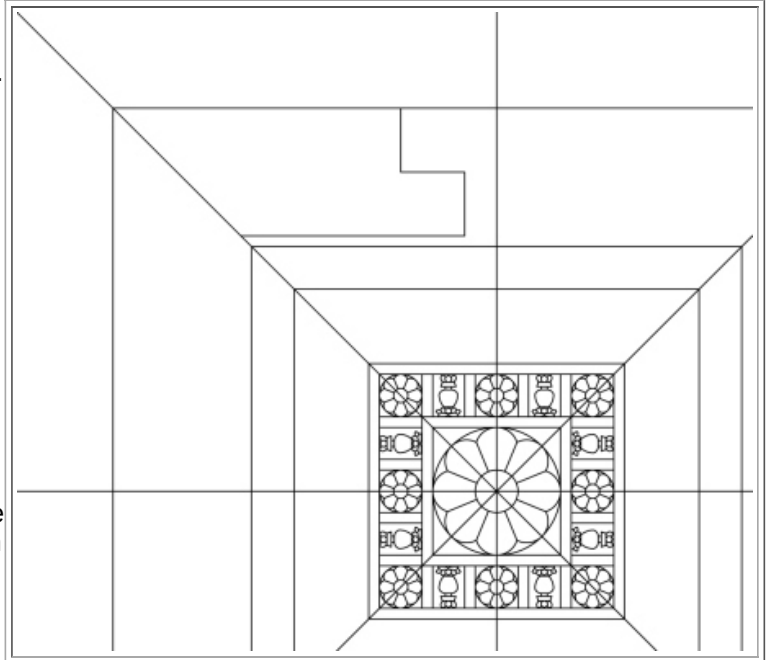
Lines now need to be drawn in the 4 mu space between the second and third squares beyond the lotus. For these, measure right and left of the central lines, 2, 1, 3 and 1 mu. The 4x4 square cells that are formed in the cardinal and intermediate directions need to have eight-petalled lotuses in them of 4 mu diameter. As with the main lotus, the diameter of the receptacle of each lotus is one third the overall diameter. The other cells formed between these are for flasks.

Beyond the outer of the four squares forming the beams, measure along the central lines 7 and then 4 mu, drawing further squares. These form the deity-podium of the mind palace. Then measure 1 mu further and draw another square. This is the base line (*rtsa thig*, *mūlasūtra*) of the mind palace; the base line is the inner line of the walls. Parts of this line need to be removed to form the doorway, 3 mu either side of the central line, giving the doorway a width of 6 mu.

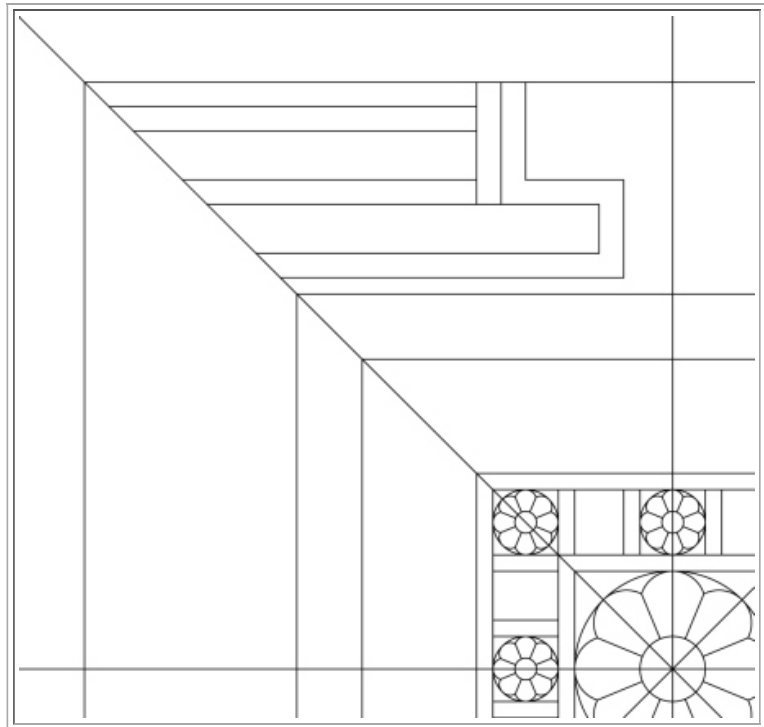


Lines are now drawn to form the porch. This consists of three sections: the porch-projection (*sgo khyud*, *niryūha*), the porch-extension (*sgo 'gram*, *kapola*) and porch-sides (*sgo logs*, *pakṣaka*). (These last two are often given the other way around. I give them here according to the use in the *Vajrāvalī*.)

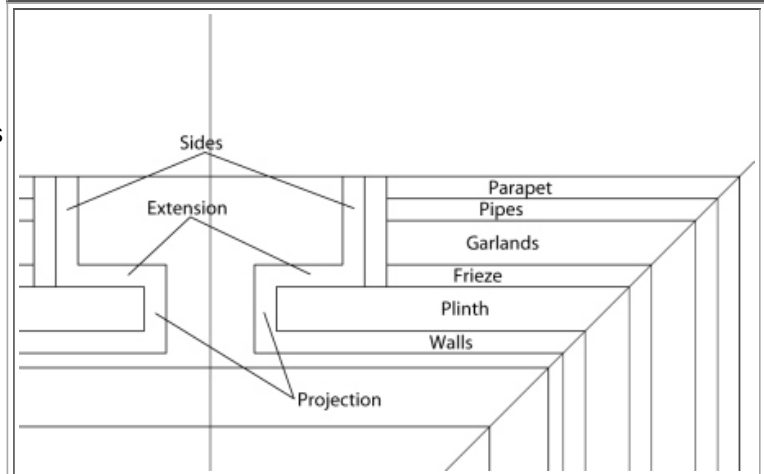
Each of the lines now to be drawn are 6 mu in length. From the point of view of the diagram, first, from the end of the base line up parallel to the central line, then horizontally away from the base line, and then again up parallel to the base line. That last line ends on the parapet line for the mind palace, drawn earlier.



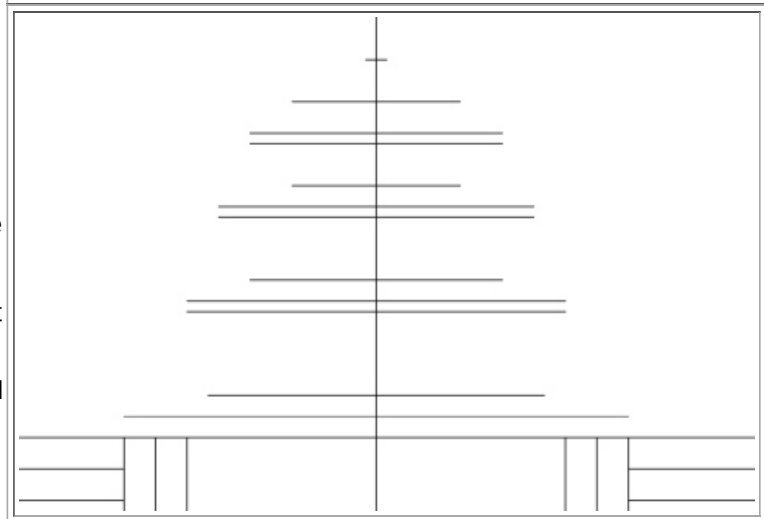
Further lines now need to be drawn between the base and parapet lines. The first should be the outer line for the walls; this is parallel to the inner wall line that has just been completed, and $1\frac{1}{2}$ mu from it. It follows the shape of the porch components. Next to the porch-sides are the pillars for the toran. These are $1\frac{1}{2}$ mu wide and $7\frac{1}{2}$ mu in length. For each of these, just one line needs to be drawn parallel to the central line. Further lines then need to be drawn parallel to the base line. Measuring up from the outer wall line: 3 , $1\frac{1}{2}$, 3 and $1\frac{1}{2}$ mu.



These form the spaces for the plinth (*stegs bu*, *vedika*), jewelled frieze (*rin po che'i pha gu*, *ratnapaṭṭika*), garlands and drops (*dra ba dang dra ba phyed pa*, *hārārdha hāra*), pipes (*rin chen shar bu*, *bakulī*) and parapet (*mda' yab*, *kramaśīrṣa*). The walls are actually more complicated than described here, consisting of parallel walls. This is described later in the *Vajrāvalī* and not mentioned at this point in the description.



The toran (*rta babs*, *toraṇa*) extends beyond the parapet line for 3 DU, and consists of three sections, or stages. Abhaya describes them by giving the size of each stage, in height, in the direction of the central line. Each stage consists of a supporting beam, railing (*myos pa srung ba*, *mattavāraṇa*) and four pillars. The first stage has a height of 6 mu, the second $4\frac{1}{2}$ and the third $3\frac{1}{2}$. On the very top is the roofing and flask top, with a height each of 2 mu. Added together, this gives a total of 18 mu, or, 3 DU.



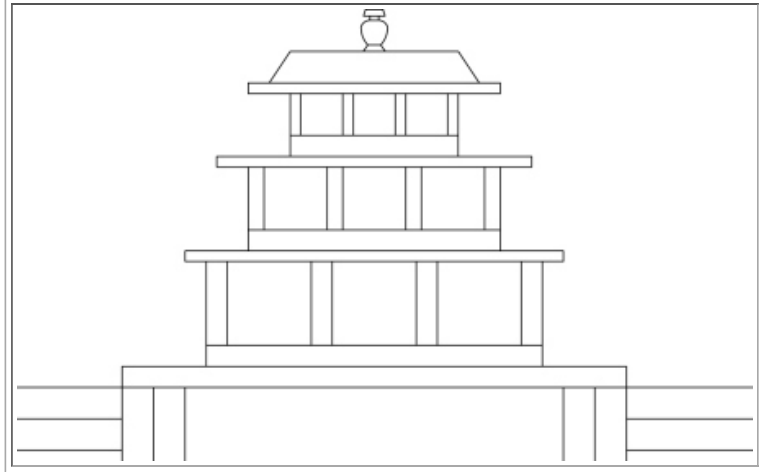
A total of 12 horizontal lines need to be drawn of varying lengths. For these, measure out from the parapet line: 1, 1, $4\frac{1}{2}$, 1, 3, $1\frac{1}{2}$, 1, 2, $1\frac{1}{2}$, $1\frac{1}{2}$ and 2 mu. These lines extend either side of the central line, respectively: 12, 8, 9, 9, 6, $7\frac{1}{2}$, $7\frac{1}{2}$, 4, 6, 6, and 4 mu.

There are only eleven in that last list. The last line is for the top of the flask that sits on top of the toran. Banda Gelek gives this a length of 2 mu, as does Tāranātha. But neither the *Vajrāvalī* nor the

Vimalaprabhā give any length for this line. Sādhuputra (author of one of the few surviving Indian maṇḍala rituals) also does not give this in his maṇḍala description. So where does this come from? In one text Dolpopa gives no length for this line, and yet in another he gives it as 2 mu. This maybe where this idea originated. But why?

The Vimalaprabhā, as with the description above and most other texts, first gives the positions of these horizontal lines and then their lengths. When describing the lengths it simply says "above this, by two, is the flask". The flask is 2 mu in height, and I would read this as simply a restatement of that height, but it is possible that Dolpopa or others took this to be an indication of the length of the line for the top of the flask. All Jonang writers and many others have subsequently followed this.

However, I am yet to see a maṇḍala that actually uses this size for the top of a flask, with the width at the top the same as its height; this would seem quite unappealing. For this reason, in the diagrams here the top of the flask is given an arbitrary width, starting at 1 mu in the diagram to the right.



The next step is to draw vertical lines to complete the structure. Counting from the bottom, the first line defines the main beam of the first stage, and vertical lines need to be drawn from its ends, down to the line below it – the parapet line. The same is done with the second line which defines the railing. Between lines two and three, eight vertical lines are drawn for the pillars, each of which is 1 mu wide, creating three cells for offering goddesses, each 4 mu across. A similar process is followed for the second stage, with the pillars each being $\frac{3}{4}$ mu wide and the cells 3 mu across. And again for the top stage, with the pillars each $\frac{1}{2}$ mu in width and the cells 2 mu across.

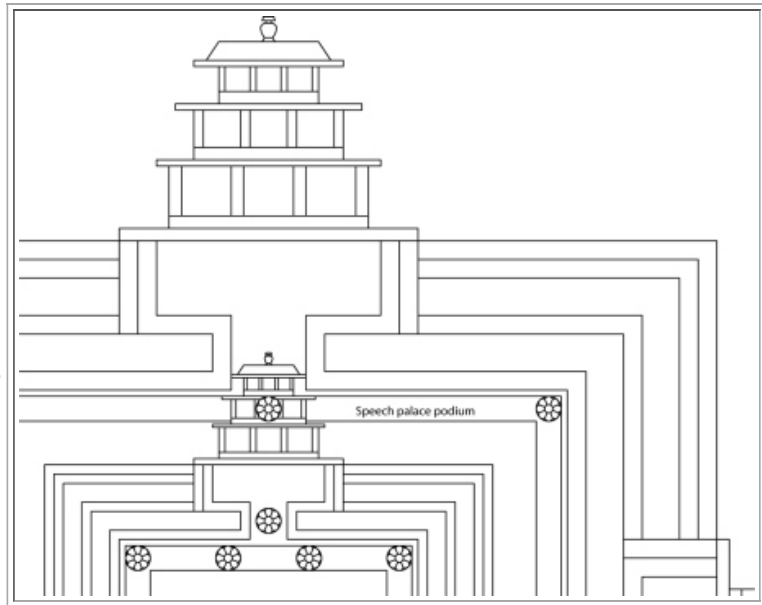
Vertical lines are then drawn between the ends of lines nine and ten to form the beam for the roof, and lines are drawn from the ends of line eleven to points on line ten, 5 mu either side of the central line, to form the roof. Finally, the flask is drawn on top, with victory banners either side.

During this description, Abhaya adds some further details. Beside the outer pillars of the first stage are (SG: white) elephants supporting (SG: blue) lions that support the beam of the second stage. Similarly for the second and third stages, beside the outer pillars are śālabhañjikā supporting the beams above. In the Tibetan translation, these are simply called goddesses; the specific term śālabhañjikā is used in both the Vajrāvalī and the Vimalaprabhā. Many Tibetan writers and artists seem to have been confused by these, even suggesting that they can be male figures. They are a very common motif in Indian architecture. The figure usually has one leg crossed in front or behind the other, and with one hand bending down the branch of a tree. The Buddha's mother is represented in a very similar pose in depictions of the birth scene. The final details given for the toran are that from the ends of all beams hang bells, chowries, mirrors, victory banner and flags.



Perhaps the most famous of all śālabhañjikā, supporting a toran beam at the great stupa of Sanchi.

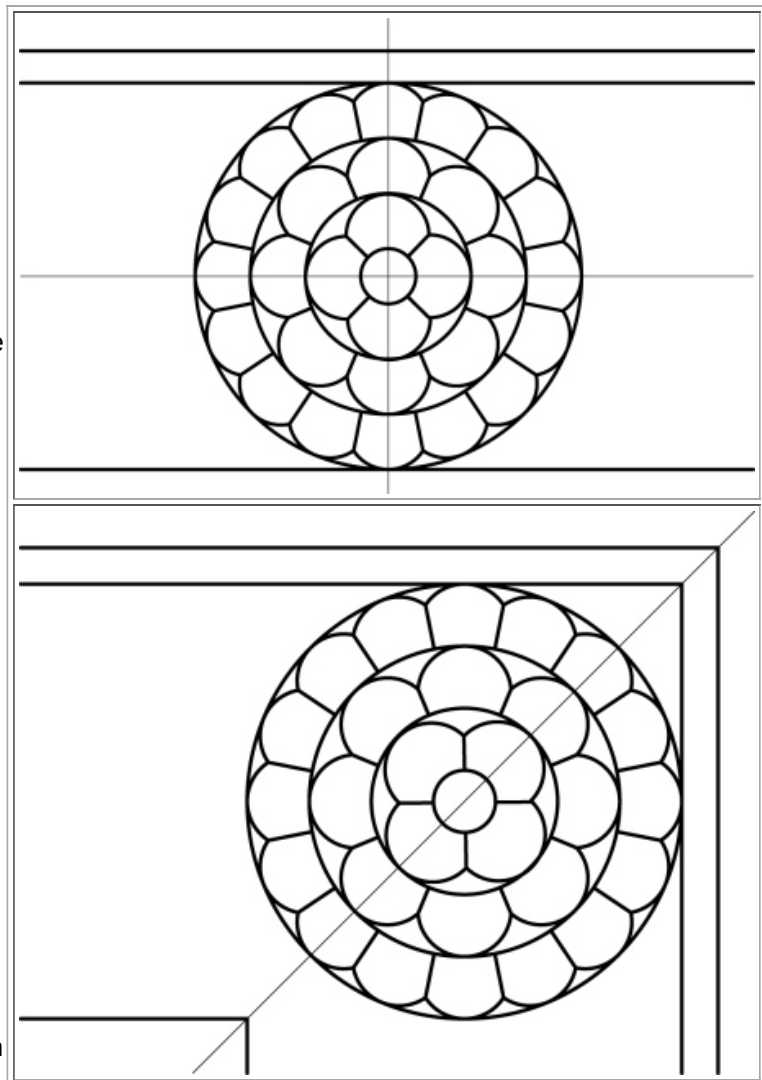
The construction now starts for the speech palace. The base line for the speech palace is 12 mu beyond the parapet line of the mind palace, and the dimensions of all the wall components from the base line up to and including the toran are all twice those of the mind palace. A deity-podium is needed between the mind parapet line and the speech base line. For this, just as with the mind palace, measure out from the mind parapet line 7 and then 4 mu, drawing squares. These form the podium, with a 1 mu narrow gap between the outer edge of the podium and the speech base line. This podium needs eight lotuses, each of diameter 4 mu and with eight petals. These are in the four corners and in the middle of each side of the podium, coinciding with the middle cell of the middle stage of each of the mind palace torans. The middle cells are 3 x 3 mu in size, and as the lotuses need to have their diameter of 4 mu, sections of the pillars and railing of the toran are cut away to accommodate the lotus.



The body palace is next. The base line for the body palace is 24 mu beyond the speech parapet line, and the dimensions of all the wall components from the base line up to and including the toran are all twice those of the speech palace.

As with the speech palace, a deity-podium is needed within the body base line. For this, measure out 11 and then 12 mu, drawing square construction lines for the podium. This leaves a small gap of 1 mu between the outer edge of the podium and the body base line. The podium needs 12 lotuses, each of diameter 12 mu. Four of these are in the four corners, and the other eight are in pairs, right and left of the doorways – right and left of the speech toran, which is drawn over this body palace deity podium. None of my Vajrāvalī source texts give an exact position for these lotuses, but in his general Kālacakra maṇḍala description, Banda Gelek says they are $27 \frac{2}{3}$ mu right and left of the central lines. That is how they are shown in the diagrams here.

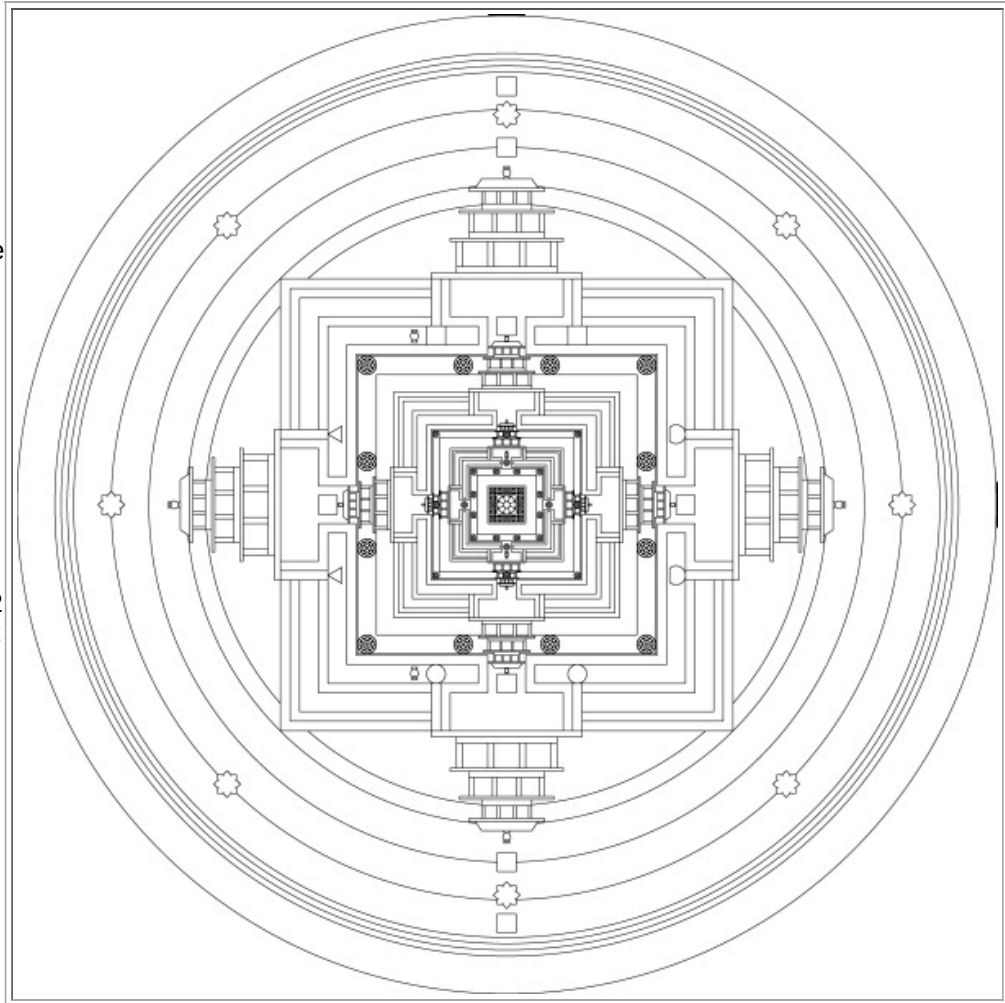
These lotuses have three bands, or rings, of petals: the inner band has four, the middle eight and the outer 16, a total of 28. The bands are all equal in width, and that width is the same as the diameter of the receptacle. In order to draw these, a diameter of the lotus is divided into seven equal parts, and three circles drawn, centred on the middle point and passing through the others. An important point to note is that in each ring of petals, one petal needs to be in a line with the centre of the maṇḍala. For those on the corners of the podium, this simply means that in each ring one petal must lie centred over the diagonal line. With the other lotuses, the same applies, but they are not turned through small



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angles in order to line up with the centre of the maṇḍala. Instead, they are drawn so that the petals line up parallel to the central line. This should be clear in the diagrams on the right.

In the middle of the doorways of the body palace, just beyond the tips of the flasks on the speech palace torans, are drawn the chariots for the protective deities. These are 12 mu in size, and are usually interpreted as being square, although they are often drawn as rectangular, with the longer side 12 mu. Another point made by Abhaya is that the tips of the body toran flasks reach to the middle of the water perimeter; the perimeters are to be drawn next. Points like this are often made in the descriptions for maṇḍala drawing,



often as a check that drawings have been made accurately. Another common point made regarding the body torans is that the centre point of top line of the railing on the top stage of the toran should coincide with the end of the central line, exactly at the circle that is the inner boundary of the perimeters, which has a radius of 192 mu. (As described earlier, strictly speaking, the central lines end at that point, although in practice they are often drawn longer.)

From that circle, we then measure out: 12, 24, 24, 24, 12 and 24 mu. The bands formed are for the perimeters of earth, water, fire, wind, space and the garland of light (*'od zer gyi phreng ba*, vajra-fire in the Vimalaprabhā). The final outer circle gives the maṇḍala an overall diameter of 624 mu.

This completes the construction for the maṇḍala. Other details are needed, but these are described in another section of the Vajrāvalī, together with the colours.

Colours

The ground that is within the two vajra-garlands of the mind maṇḍala – the square sets of beams – is all black, as is the eastern ground in all three palaces. (SG: The ground within the tathāgata-dais and the pillars is blue.) The eastern ground refers to the eastern ground between the diagonal lines. (This may seem obvious, but other definitions have apparently been in use.) The ground of all three palaces is in the south, red, in the west, yellow and in the north, white. The ground includes the small gaps between the podium and the base line of the next palace.

Of the wall structures, the plinths are white, the frieze is red and adorned with various coloured jewels (often in the shape of deep blue triangles, red semi-circles, yellow squares and white circles). The garlands and drops are white on a black background. This is also the case with the pipes (drawn like upside-down bottles) and the merlons of the parapet; the latter are designed like half lotus petals (that is the description used in Tibetan, but in practice they look nothing like lotus petals). The main toran pillars,

either side of the doorways, are yellow. (SG: The corners of the walls are adorned with crossed-vajras that are white.)

The central lotus in the mind palace is green. The inner vajra-garland and the pillars are all black. The pillars in the east are decorated with black swords, in the south with red jewels, in the west with yellow wheels, and in the north with white lotuses. The eight flasks are drawn in the 3 mu-wide cells between the pillars, and are coloured white, each seated on a lotus and topped by a lotus.

The outer vajra-garland is green. The deity-podium is white (the current description is for the mind palace, but this is true of all three palaces). The walls of the mind palace are triple, from the inside, white, red and black. (SG: Has green instead of black. This is presumably an error; other authors, such as Ngawang Lozang Choden, *ngag dbang blo bzang chos ldan*, author of the main Vajrāvalī empowerment text used in the Gelug tradition, have black.) There are small gaps between these, and in order to draw them, the $1\frac{1}{2}$ mu space drawn for the walls is divided into eight equal parts. (Abhaya actually says nine, but this seems just to be a different grammar for making the division; it only works if you divide into eight parts.) Each wall takes up two of the narrow bands thus formed, and this leaves two gaps of one band in width between the walls. It is not stated in the texts, and it would be a very small detail to paint, but the gaps between the walls should be the colour of the directions; the colour of the ground.

The sides of the door-extensions are adorned with vajras and jewels.

All twelve torans are multi-coloured. (SG: The lower beams of all toran are green, the railings are red and all pillars yellow. The beams for the upper two stages are blue, the beam for the roof is red and the roof itself is blue. For the backgrounds of the cells of the torans: all the lower cells are blue; all the upper cells are green; for the middle cells: in the mind toran they are blue-green, for the speech, the middle cells are green and right and left red, the body all are white. Each mind and speech toran has only drawings of jewelled garlands as ornaments. The flasks on top of all torans are red, each topped with a jewel. For the other toran, in the three middle cells are offering goddesses, carrying offerings and in the three upper, simply jewel garlands.)

The speech podium is white. The lotuses on it do not have sun or moon disks for seats. The lotuses in the cardinal directions are red, and those in the corners are white. The walls of the palace are five-fold, and these are, from the inside, green, black, red, white and yellow. The space for the walls is divided into 14 equal bands, and as before, two bands are allotted to each wall, leaving four single bands between the walls.

The podium and the walls are coloured in the body palace the same as the speech palace. The lotuses on the body podium are also without moon or sun seats, and those in the corners are white and all the others red.

On the body palace plinths, underneath the main toran pillars, are pairs of seats (of the four elements) for the nāgas. These each have a size of 12 mu. In the east these are black, circular and have victory banners drawn on them; in the south they are red, triangular and with svastikas; in the west they are yellow, square and with vajras; and, in the north they are white, semi-circular and with lotuses. Many Gelug painted maṇḍalas will show the element disks in the north and east the other way around. This is presumably because they are given this way in the sādhana by the 7th Dalai Lama. Associating the semi-circular design with wind and circular with water is the normal Buddhist tradition. The reverse is particular to Kālacakra. Most writers state that it does not matter which method is followed.

To the right of the nāga seat right of the eastern doorway is a flask of nectar representing the above nāga, Jaya; to the left of the square nāga seat left of the western doorway is another flask of nectar representing the below nāga, Vijaya.

In the central cell of the lower stage of the eastern body toran is a black dharmacakra; right and left of this are black male and female deer (this is usually interpreted as suggesting they are in the right and left cells, but some consider all three to be in the central cell). In a similar position in the south is a red "excellent flask", with a conch to the right and a red lotus to the left. In the west is a yellow bodhi-tree, with a kinnara and kinnarī. In the north is a white great drum with a club and hammer.

The ground between the parapet and the inner edge of the perimeter of earth is black, and should have many various offerings drawn on it. This area is often referred to as the offering ground (*mchod pa'i sa gzhi*).

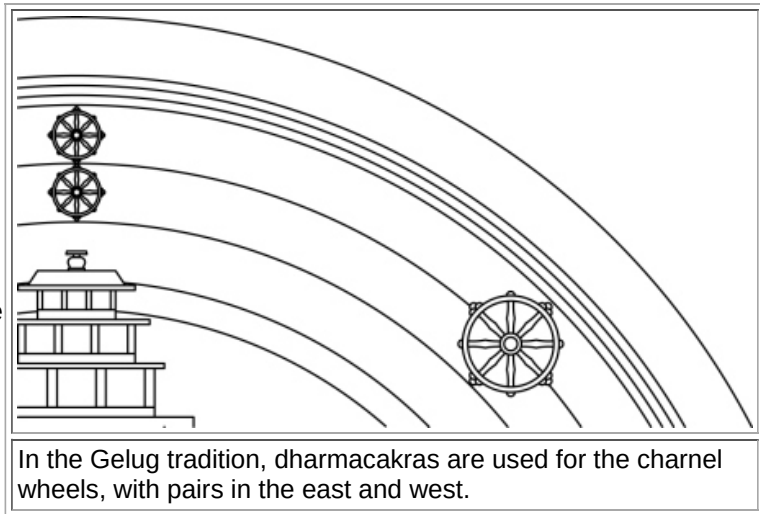
The earth perimeter is yellow, and in its NE is the disk of the rising full Moon, with a diameter of 12 mu. In the SW is the disk of the setting Sun, also 12 mu. At the exact NE and SW positions, the corners of the

body parapet overlap the earth perimeter, and so the Sun and Moon have to be drawn just to one side of these corners.

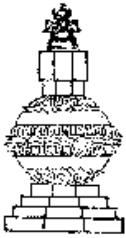
The water perimeter is white (SG: with blue designs), the fire perimeter red, the wind perimeter black. On the junction of the fire and wind perimeters are eight eight-spoked charnel ground wheels with diameters of 12 mu. Those in the cardinal directions are red and in the intermediate directions white. (The charnel ground wheels (*dur khrod kyi 'khor lo*) are interpreted differently.

Some draw these as dharmacakras and other as wrathful weapon wheels; the latter seems more appropriate for the charnel ground seat of a wrathful goddess and so has been used in the main diagrams given here.) Beyond the eastern wheel is a disk of emptiness, and another beyond the western wheel. (These were later interpreted as the elements of space and awareness, respectively.) All ten of these have curved knives as emblems. Also, draw many different emblems in the wind perimeter. (This is a brief reference to the perimeter beings; later Tibetan traditions of these are [described here](#).)

The space perimeter is black. Further circles need to be drawn in this to create three bands: 3, 6 and 3 mu wide. The outer two of these are the actual space perimeter and the middle band is used to draw the green vajra-garland. The blazing light of the outer perimeter is of five colours (SG: and it swirls to the right, clock-wise).



The [next page](#) describes the emblems of the deities.



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